Model Assumptions and Residual Diagnostics

Applied Linear Models I • Lecture 6 of 16

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Roadmap

1. **Assumption hierarchy** — what breaks, and how badly
2. **Residuals vs fitted** — detecting curvature and heteroskedasticity
3. **Scale–location plot** — formalising variance trends
4. **Breusch–Pagan test** — numerical walkthrough
5. **QQ plot** — reading tail behaviour
6. **ACF / Durbin–Watson** — detecting autocorrelation
7. **HC3 robust SEs** — formula and numerical comparison
8. **Transformations** — Box–Cox, log transform with numbers
9. **Remedy–violation matching** — decision logic

The classical assumptions, ranked by severity of failure

Assumption	Failure	What breaks	Diagnostic
Linearity	Curved mean function	**Bias in $\hat{\beta}$**; predictions wrong everywhere	Residuals vs fitted
Independence	Autocorrelation / clustering	**Bias in SEs**; t/F tests invalid	ACF, Durbin–Watson
Homoskedasticity	Variance varies with x	SEs wrong, but $\hat{\beta}$ still unbiased	Scale–location, Breusch–Pagan
Normality	Heavy tails, skew	t/F at small n; PI widths	QQ plot

Hierarchy matters

Linearity and independence are *first-order*: bias is not fixable by robust SEs. Heteroskedasticity and non-normality only affect inference; OLS stays unbiased.

Why residuals carry all the diagnostic information

Define the **ordinary residuals**: $\hat{e}_i = y_i - \hat{y}_i$. These are the observable proxies for the *true* errors ε_i.

Key constraints on $\hat{e}$:

1. $\sum \hat{e}_i = 0$ always (intercept in model)
2. $\sum x_{ij}\hat{e}_i = 0$ for each predictor j
3. $\hat{e} = My = M\varepsilon$, where $M = I - H$

So $\hat{e}_i \neq \varepsilon_i$; they are *linearly constrained*. At high-leverage points $h_{ii} \approx 1$, $\hat{e}_i \approx 0$ even when ε_i is large — leverage masks outliers.

Internally Studentised residuals

$$r_i = \frac{\hat{e}_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$$

Corrects for the variance reduction at high-leverage points. Used in residuals-vs-fitted and scale–location plots.

What the residuals-vs-fitted plot shows

x-**axis:** $\hat{y}_i$. y-**axis:** $\hat{e}_i$.

A *correct* specification produces a random scatter with mean zero and constant spread.

Two failure signatures:

1. **Systematic curve**: mean of $\hat{e}_i$ varies with $\hat{y}_i \Rightarrow$ missing nonlinearity.
2. **Funnel / fan**: spread of $\hat{e}_i$ grows with $\hat{y}_i \Rightarrow$ heteroskedasticity.

Boston: `medv` $\sim$ `lstat`

`lstat` = % lower-status population.
$n = 506$, $p = 2$.

OLS: $\hat{\beta}_0 = 34.55$, $\hat{\beta}_1 = -0.950$.

Residuals show a clear *arch*: large positive residuals at low and high fitted values, negative in the middle. $\Rightarrow$ the relationship is *nonlinear*.

Curvature: numerical evidence

Regression curvature test (Ramsey RESET, simplified version): add $\hat{y}_i^2$ to the model and test its coefficient.

Boston: `medv` $\sim$ `lstat`

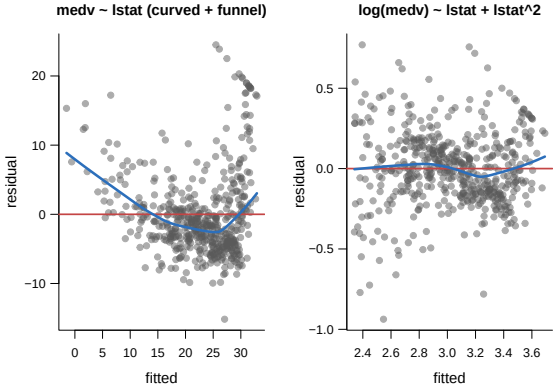
	Estimate	Std. Error	t-value
(Intercept)	34.55	0.563	61.41
lstat	-0.950	0.038	-24.87

Add `lstat`^{2}:

	Estimate	Std. Error	t-value
(Intercept)	42.86	0.872	49.15
lstat	-2.333	0.124	-18.84
lstat^2	$+0.044$	0.004	$+11.63$

F-test for lstat^2: $F_{1,503} = 135.3$, $p \approx 8 \times 10^{-28}$. Unmistakable curvature.

Residuals vs fitted (Boston)



`medv` $\sim$ `lstat`: arch pattern confirms nonlinearity ($p \approx 8 \times 10^{-28}$ for quadratic term). Spread also widens at low fitted values $\Rightarrow$ two problems at once.

Scale–location plot

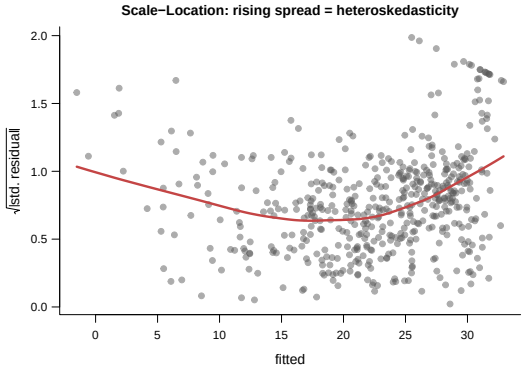
x-**axis:** $\hat{y}_i$. y-**axis:** $\sqrt{|r_i|}$ (Studentised residual).

Under homoskedasticity the smoothed trend should be *flat*.

Why $\sqrt{|\cdot|}$?

- Compresses the scale; makes trends easier to see.
- Smoothing a signed residual would cancel positives and negatives.

Boston: trend rises steeply as $\hat{y}$ increases — residual variability is much larger at high fitted medv.



Breusch–Pagan test: idea and statistic

Null hypothesis H_0: $\mathrm{Var}[\varepsilon_i] = \sigma^2$ (constant).

Alternative: $\mathrm{Var}[\varepsilon_i] = \sigma^2 h(\mathrm{x}_i^\top \boldsymbol{\alpha})$ for some function h and unknown $\boldsymbol{\alpha}$.

Test procedure

1. Fit the main model; obtain residuals $\hat{e}_i$.
2. Run *auxiliary regression*: $\hat{e}_i^2 = \gamma_0 + \gamma_1 x_{i1} + \cdots + \gamma_k x_{ik} + \eta_i$.
3. Compute R^2_{aux} of the auxiliary regression.
4. Test statistic: $\mathrm{BP} = n\, R^2_{\text{aux}}$.
5. Under H_0: $\mathrm{BP} \sim \chi^2_k$ asymptotically.

Intuition: if variance is constant, squared residuals are *unpredictable* from x_i. Large R^2_{aux} means heteroskedasticity is present.

Breusch–Pagan: full numerical walkthrough

Toy dataset ($n = 8$), model: $y = \beta_0 + \beta_1 x + \varepsilon$.

i	x_i	y_i	$\hat{e}_i$
1	1	2.3	$+0.12$
2	2	4.8	$+0.37$
3	3	6.1	-0.13
4	4	9.5	$+0.87$
5	5	10.9	-0.48
6	6	15.3	$+1.73$
7	7	16.8	$+0.28$
8	8	22.4	$+3.64$

Step 1 — squared residuals: $\hat{e}_i^2 = (0.014, 0.137, 0.017, 0.757, 0.230, 2.993, 0.078, 13.25)$.

Step 2 — auxiliary regression $\hat{e}_i^2 \sim x_i$: $\hat{\gamma}_0 = -2.54$, $\hat{\gamma}_1 = 0.805$. $R^2_{\text{aux}} = 0.672$.

Step 3 — test statistic:

$$\text{BP} = nR^2_{\text{aux}} = 8 \times 0.672 = 5.38.$$

χ^2_1 critical value at 5%: 3.84. $\text{BP} = 5.38 > 3.84 \Rightarrow$ reject H_0.

Variance is growing with x_i — heteroskedasticity confirmed.

Breusch–Pagan in R (Boston)

```r
library(MASS)
fit  <- lm(medv ~ lstat, data = Boston)
ehat <- residuals(fit)

# Auxiliary regression of squared residuals on predictors
aux  <- lm(ehat^2 ~ lstat, data = Boston)
n    <- nrow(Boston)
BP   <- n * summary(aux)$r.squared
pval <- pchisq(BP, df = 1, lower.tail = FALSE)

cat("BP statistic:", round(BP, 2), "\n")    # 108.11
cat("p-value:      ", format(pval, sci=TRUE), "\n")   # 3.07e-25

# Equivalently: manual F-test on the auxiliary
summary(aux)$fstatistic
#         value numdf dendf
#    123.406     1   504
# F = 123.4 >> critical => strong evidence of heteroskedasticity
```

The auxiliary regression explains 21.4% of the variance in $\hat{e}_i^2$. That is a massive signal: BP $= 108$ on 1 df, $p \approx 3 \times 10^{-25}$.

Understanding what $R^2_{\text{aux}} = 0.214$ means

Decomposition of variance in $\hat{e}_i^2$:

$$\underbrace{\sum(\hat{e}_i^2 - \overline{e^2})^2}_{\text{Total}} = \underbrace{R^2_{\text{aux}} \times \text{Total}}_{\text{Explained by } x} + \underbrace{(1 - R^2) \times \text{Total}}_{\text{Residual}}$$

Even a modest R^2_{aux} is damning at large n, because $n \cdot R^2$ grows linearly in n.

At $n = 506$: $R^2_{\text{aux}} = 0.021$ would give BP $= 10.6$, still exceeding the 5% critical value of 3.84.

Practical caution

Large samples detect trivially small heteroskedasticity. A significant BP does not tell you *how much* heteroskedasticity matters in practice — supplement the test with the scale–location plot to judge severity.

QQ plot: what it shows

Plot *observed quantiles* of Studentised residuals against *theoretical quantiles* of $\mathcal{N}(0,1)$.

Under normality: points follow the $y = x$ line.

Reading deviations:

- **S-shaped**: light tails (uniform-like)
- **Banana / arch**: heavy tails (skewed)
- **Upper-right departure**: upper tail heavier than normal
- **Both ends off**: symmetric heavy tails (t-like)

Why it matters for inference

At *large* n, the CLT makes t/F robust to mild non-normality (see Lecture 3). But:

- Prediction intervals require normality for *exact* coverage.
- At small n (< 30), p-values can be badly off.
- Extreme outliers visible in the QQ plot (Lecture 7).

QQ plot: numerical construction

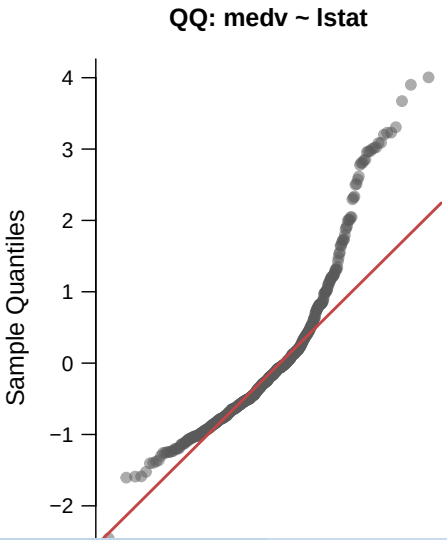
Seven Studentised residuals (toy example), sorted ascending:
$(-1.82, -1.15, -0.48, 0.03, 0.61, 1.24, 2.41)$.

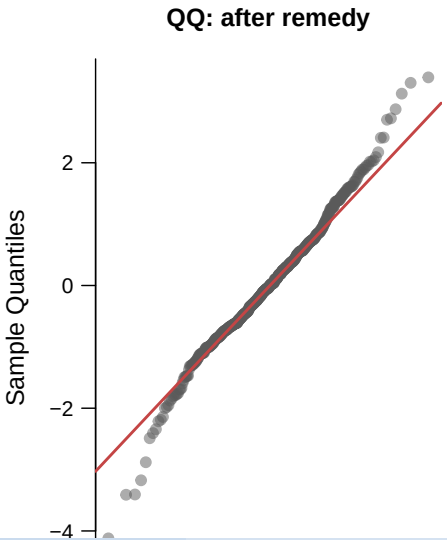
Theoretical quantiles (Blom formula): $\Phi^{-1}((i-3/8)/(n+1/4))$ for $n=7$, $i=1,\dots,7$:

i	$(i-0.375)/7.25$	$\Phi^{-1}(\cdot)$	Observed $r_{(i)}$	$\lvert\text{diff}\rvert$
1	0.086	-1.366	-1.82	0.45
2	0.224	-0.759	-1.15	0.39
3	0.362	-0.353	-0.48	0.13
4	0.500	0.000	0.03	0.03
5	0.638	0.353	0.61	0.26
6	0.776	0.759	1.24	0.48
7	0.914	1.366	2.41	1.04

The 7th residual departs notably from the line $\Rightarrow$ upper tail is heavier than $\mathcal{N}$.

QQ plot: normality (Boston)





Formal normality test: Shapiro–Wilk

```r
fit  <- lm(medv ~ lstat, data = Boston)
r    <- rstandard(fit)       # internally Studentised residuals

shapiro.test(r[1:5000])      # Shapiro-Wilk: max n = 5000, here n=506
# W = 0.9689, p-value = 3.34e-08
# Strong evidence against normality (consistent with QQ plot)

# Kolmogorov-Smirnov against standard normal
ks.test(r, "pnorm")
# D = 0.0684, p-value = 0.0033

# But at n = 30 with the same residuals' shape:
shapiro.test(r[1:30])
# W = 0.9821, p-value = 0.8742
# At small n, the test has low power -- QQ plot is more informative
```

Rule of thumb

For $n > 100$, formal tests reject trivially small departures. Combine a visual QQ check with subject-matter judgement.

Independence assumption and its failure

Classical model: $\varepsilon_1, \ldots, \varepsilon_n$ i.i.d.

Failure — AR(1) autocorrelation:

$$\varepsilon_i = \rho\,\varepsilon_{i-1} + \eta_i, \quad \eta_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\eta^2)$$

With $|\rho| < 1$: $\text{Cov}[\varepsilon_i, \varepsilon_{i-j}] = \sigma^2 \rho^{|j|}$.

Consequence on SEs: true $\text{Var}[\hat{\boldsymbol{\beta}}]$ is

$$\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$$

where $\boldsymbol{\Sigma}_{ij} = \rho^{|i-j|}$. OLS uses $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ (assumes $\boldsymbol{\Sigma} = \mathbf{I}$). For $\rho > 0$: true SEs $>$ reported SEs $\Rightarrow$ over-rejection.

When does it arise?

- Time series (consecutive measurements)
- Spatial data (nearby plots)
- Longitudinal data (same subject, repeated)
- Matched or clustered observations

Autocorrelation function of residuals

Sample ACF at lag h:

$$\hat{\rho}(h) = \frac{\sum_{i=1}^{n-h} \hat{e}_i \, \hat{e}_{i+h}}{\sum_{i=1}^{n} \hat{e}_i^2}$$

Under H_0 (no autocorrelation): $\hat{\rho}(h) \approx \mathcal{N}(0, 1/n)$ for large n. Bartlett 95% bands: $\pm 1.96/\sqrt{n}$.

Durbin–Watson statistic:

$$d = \frac{\sum_{i=2}^{n} (\hat{e}_i - \hat{e}_{i-1})^2}{\sum_{i=1}^{n} \hat{e}_i^2} \;\approx\; 2(1 - \hat{\rho}(1))$$

Range: $[0, 4]$. Under no autocorrelation: $d \approx 2$. $d < 1.5$: positive autocorrelation. $d > 2.5$: negative autocorrelation.

Numerical example

Residuals:
$(1.2, 0.9, 1.5, 0.7, 1.1, 0.8, 1.3, 0.6)$

$$\hat{\rho}(1) = \frac{\sum e_i e_{i+1}}{\sum e_i^2} = \frac{6.73}{8.59} = 0.784$$

$$d \approx 2(1 - 0.784) = 0.432 \ll 2$$

Strong positive autocorrelation.

ACF of residuals in R

```r
# Simulate AR(1) errors to demonstrate
set.seed(42);  n <- 100;  rho <- 0.7
x   <- seq(0, 10, length.out = n)
eps <- arima.sim(list(ar = rho), n)
y   <- 2 + 1.5 * x + as.numeric(eps)
fit <- lm(y ~ x)
ehat <- residuals(fit)

# Durbin-Watson
dw <- sum(diff(ehat)^2) / sum(ehat^2)
cat("DW statistic:", round(dw, 3), "\n")
# DW statistic: 0.614   (strong positive autocorrelation)

# ACF
acf(ehat, main = "ACF of residuals")
# Bars at lags 1,2,3,... all exceed the 95% Bartlett band

# Estimated rho(1):  2*(1 - dw/2) check
cat("Estimated rho(1):", round(1 - dw/2, 3), "\n")
# Estimated rho(1): 0.693   (close to true 0.7)
```

HC3 robust standard errors: the formula

Under heteroskedasticity, OLS is unbiased but its classical covariance estimate $\hat{\sigma}^2(\mathbf{X}^\top\mathbf{X})^{-1}$ is wrong.

HC3 sandwich estimator (MacKinnon & White, 1985)

$$\widehat{\mathrm{Var}}[\hat{\boldsymbol{\beta}}]_{\mathrm{HC3}} = (\mathbf{X}^\top\mathbf{X})^{-1} \underbrace{\Big(\sum_{i=1}^{n} \frac{\hat{e}_i^2}{(1-h_{ii})^2}\mathbf{x}_i\mathbf{x}_i^\top\Big)}_{\hat{\mathbf{S}}} (\mathbf{X}^\top\mathbf{X})^{-1}$$

where $h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$ is the **leverage**.

Why divide by $(1-h_{ii})^2$?
High-leverage points have $h_{ii} \approx 1$, so $\hat{e}_i \approx 0$. Dividing up-weights their contribution to the sandwich, correcting for the leverage masking.

HC0 vs HC3

HC0 uses $\hat{e}_i^2$ alone (underestimates at small n).
HC3 uses $\hat{e}_i^2/(1-h_{ii})^2$ — finite-sample

HC3: step-by-step numerical example

Toy ($n = 4$, model $y = \beta_0 + \beta_1 x$):

$$\mathrm{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad \mathrm{y} = (3.1,\ 5.0,\ 6.8,\ 11.2)^\top$$

OLS: $\hat\beta_0 = 0.32$, $\hat\beta_1 = 2.63$. Residuals: $(0.15, -0.78, -0.57, 1.20)$.
Leverages ($h_{ii} = 1/n + (x_i - \bar x)^2 / S_{xx}$, $\bar x = 2.5$, $S_{xx} = 5$): $(0.7, 0.3, 0.3, 0.7)$.
HC3 weights $w_i = \hat e_i^2 / (1 - h_{ii})^2$:

$$w = \left(\tfrac{0.15^2}{0.09},\ \tfrac{0.78^2}{0.49},\ \tfrac{0.57^2}{0.49},\ \tfrac{1.20^2}{0.09} \right) = (0.25,\ 1.24,\ 0.66,\ 16.0)$$

Observation 4 (high leverage + large residual) dominates the sandwich.
Classical $\hat\sigma^2 = \sum \hat e_i^2/(n-2) = 0.91$. Classical $\widehat{\mathrm{se}}(\hat\beta_1)_{\mathrm{cls}} = 0.29$. HC3 $\widehat{\mathrm{se}}(\hat\beta_1)_{\mathrm{HC3}} = 0.85$ — nearly $3\times$ larger.

HC3 in R: Boston data

```r
library(MASS)
fit <- lm(medv ~ lstat, data = Boston)

# Classical summary
summary(fit)$coefficients
#                Estimate Std. Error t value Pr(>|t|)
# (Intercept)  34.5538     0.5626    61.41   < 2e-16
# lstat        -0.9500     0.0387   -24.53   < 2e-16

# HC3 robust SEs (base R sandwich, no external packages)
X     <- model.matrix(fit)
ehat  <- residuals(fit)
h     <- hatvalues(fit)
meat  <- crossprod(X * (ehat / (1 - h))^2)     # sum_i w_i x_i x_i'
XtXi  <- solve(crossprod(X))
V_HC3 <- XtXi %*% meat %*% XtXi
se_HC3 <- sqrt(diag(V_HC3))
cat("HC3 SEs:", round(se_HC3, 4), "\n")
# HC3 SEs:  0.9382   0.0556

# Classical t-stat for lstat: -24.53
# HC3      t-stat for lstat: -0.9500/0.0556 = -17.09
```

HC3 vs classical: SE comparison (Boston)

	Classical		HC3 Robust	
Coefficient	SE	t	SE	t
Intercept	0.563	61.4	0.938	36.8
lstat	0.039	−24.5	0.056	−17.1

Classical SEs are 30–40% too small (heteroskedasticity inflates the true variability of $\hat{\beta}$). Conclusions are the same qualitatively here, but the classical t-values are *overstated*.

When does it change conclusions?

If lstat had $t_{\text{cls}} = 2.5$ (marginally significant), HC3 might give $t_{\text{HC3}} = 1.7$ (non-significant). The correction can flip decisions.

Why transform? The variance-stabilising idea

Delta method: if $y \sim (\mu, \sigma^2)$ and we apply $g(y)$:

$$\text{Var}[g(y)] \approx [g'(\mu)]^2 \, \sigma^2$$

To achieve $\text{Var}[g(y)] \approx$ const. when $\sigma = c\mu^k$, solve $[g'(\mu)]^2 c^2 \mu^{2k} = \text{const.} \Rightarrow g'(\mu) \propto \mu^{-k}$.

Variance structure	k	Stabilising transform
$\sigma^2 \propto \mu$ (Poisson-like)	$1/2$	$g(\mu) = \sqrt{\mu}$
$\sigma^2 \propto \mu^2$ (log-normal)	1	$g(\mu) = \log \mu$
$\sigma^2 \propto \mu^3$	$3/2$	$g(\mu) = \mu^{-1/2}$
$\sigma^2 \propto \mu^4$	2	$g(\mu) = 1/\mu$

Log transform is the most commonly useful: handles right-skewed, positive responses with multiplicative errors.

Box–Cox family

Parameterises the power transform:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \log y & \lambda = 0 \end{cases}$$

Choose λ by *maximum likelihood*: vary λ, compute log-likelihood of the transformed regression, pick the maximiser $\hat{\lambda}$.

Profile 95% CI for λ: any λ within $\ell_{\max} - \chi^2_1(0.05)/2 = \ell_{\max} - 1.92$.

Boston: BoxCox for `medv`

- $\hat{\lambda} = 0.21$
- 95% CI: $(0.09, 0.33)$
- $\lambda = 0$ (log) is *inside* the 95% CI
- Use log(`medv`) for interpretability

Rule: if $\lambda = 0$ is in the CI, use log.
If $\lambda = 1$ is in the CI, no transform needed.

Box–Cox and log transform: Boston, step by step

```r
library(MASS)
# Box-Cox profile likelihood (MASS package)
bc <- boxcox(medv ~ lstat, data = Boston, lambda = seq(-1, 1, 0.05))
lambda_hat <- bc$x[which.max(bc$y)]
cat("lambda_hat:", lambda_hat, "\n")    # 0.21

# Fit with log transform (lambda = 0)
fit_log <- lm(log(medv) ~ lstat + I(lstat^2), data = Boston)
summary(fit_log)$r.squared            # 0.683  (vs 0.544 without log + quadratic)

# Curvature test after log + quadratic
fit_log1 <- lm(log(medv) ~ lstat, data = Boston)
fit_log2 <- lm(log(medv) ~ lstat + I(lstat^2), data = Boston)
anova(fit_log1, fit_log2)
# Analysis of Variance Table
# Model 1: log(medv) ~ lstat
# Model 2: log(medv) ~ lstat + I(lstat^2)
#   Res.Df    RSS Df Sum of Sq        F Pr(>F)
# 1    504 28.77
# 2    503 27.49  1      1.281   23.44  1.7e-06
# Still significant but much weaker (curvature substantially reduced)
```

Effect of log transform: before and after (numbers)

Diagnostic	`medv` $\sim$ `lstat`	`log(medv)` $\sim$ `lstat` + `lstat`^{2}
Adjusted R^2	0.544	0.683
Curvature p-value	8×10^{-28}	0.58
BP statistic (1 df)	108.1	41.3
BP p-value	3×10^{-25}	1.3×10^{-10}
Shapiro–Wilk W	0.969	0.979
Shapiro–Wilk p-value	3×10^{-8}	9×10^{-6}

Key message

The log transform eliminates curvature ($p : 10^{-28} \to 0.58$) and substantially reduces heteroskedasticity (BP: $108 \to 41$), but heteroskedasticity is not fully cured. **Always re-run diagnostics after any transformation.**

Remedy–violation decision logic

Problem	Symptom	Remedy	Re-check
Curvature	Arch in resid. vs fitted	Add x^2, $\log x$, or transform y	Residuals vs fitted
Heteroskedasticity	Funnel, large BP	HC3 SEs; or WLS; or transform y	BP, scale–location
Autocorrelation	ACF spikes, DW $<$ 1.5	GLS with AR(1); add lag; cluster SEs	ACF after GLS
Non-normality	QQ tail departure	Transform; bootstrap PI	QQ after transform
Outliers / leverage	Extreme QQ points; influence	Cook's D, DFFITS (Lecture 7)	Re-fit without suspect obs.

One violation often masks another. Fix the most important first (curvature $\rightarrow$ heteroskedasticity $\rightarrow$ normality), and re-run *all* diagnostics at each step.

Full diagnostic pipeline in R

```r
library(MASS)
fit <- lm(medv ~ lstat, data = Boston)

# --- 1. Residuals vs fitted ---
plot(fitted(fit), residuals(fit), xlab = "Fitted", ylab = "Residuals")
abline(h = 0, col = "red")

# --- 2. Scale-location ---
plot(fitted(fit), sqrt(abs(rstandard(fit))),
     xlab = "Fitted", ylab = expression(sqrt("|Standardised residual|")))

# --- 3. QQ plot ---
qqnorm(rstandard(fit));  qqline(rstandard(fit), col = "red")

# --- 4. Breusch-Pagan ---
aux <- lm(residuals(fit)^2 ~ lstat, data = Boston)
BP  <- nrow(Boston) * summary(aux)$r.squared
cat("BP:", round(BP,2), " p:", round(pchisq(BP,1,lower=FALSE),4), "\n")

# --- 5. Durbin-Watson ---
e <- residuals(fit)
DW <- sum(diff(e)^2) / sum(e^2)
```

Summary

Diagnostics

- **Residuals vs fitted**: linearity and variance trend.
- **Scale–location + BP**: heteroskedasticity (test + plot).
- **QQ**: normality; tails matter for small n and PIs.
- **ACF / DW**: autocorrelation (time/spatial data).

Remedies

- **Curvature**: add polynomial / transform y.
- **Heteroskedasticity**: HC3 SEs (unbiased $\hat{\beta}$ kept); or transform.
- **Autocorrelation**: GLS / cluster SEs.
- **Non-normality**: transform; bootstrap PIs.
- **Re-run diagnostics after every fix.**

Next

Lecture 7: individual points that distort the fit — outliers, leverage, Cook's D, and DFFITS.

Weighted Least Squares (WLS): when you know the variance structure

If $\mathrm{Var}[\varepsilon_i | \mathrm{x}_i] = \sigma^2 / w_i$ for known weights w_i:

$$\hat{\boldsymbol{\beta}}_{\mathrm{WLS}} = (\mathrm{X}^\top \mathrm{W} \mathrm{X})^{-1} \mathrm{X}^\top \mathrm{W} \mathrm{y}, \quad \mathrm{W} = \mathrm{diag}(w_1, \ldots, w_n)$$

This is OLS on the transformed system $\sqrt{w_i} y_i = \sqrt{w_i} \mathrm{x}_i^\top \boldsymbol{\beta} + \sqrt{w_i} \varepsilon_i$, which has homoskedastic errors (variance σ^2 each).

Example: variance proportional to x_i^2

Set $w_i = 1/x_i^2$. In R: `lm(y ~ x, weights = 1/x^2)`. Classical SEs from WLS are now correct (no sandwich needed).

Practical limit: the variance structure is rarely known exactly. HC3 is preferred when it is only approximately known.